ALGEBRA 2 · UNIT 7: RATIONAL FUNCTIONS

Mural Business

Run a mural-painting business with rational functions, work rates, and rational equations.

CP2

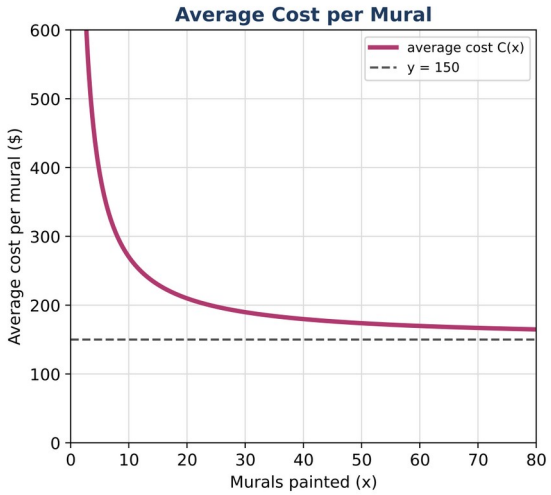
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

You and your friends are starting a mural business. Equipment costs $1,200 once, and materials cost $150 per mural. The more murals you paint, the lower your average cost per mural.

You will also schedule painters who work at different speeds and plan your travel. Every question leads to a rational function or equation.



Mural Co.

Equipment $1,200

Materials $150

per mural

Ana: 6 h per mural

Ben: 9 h per mural

Learning Goals

- I can evaluate and graph a rational function and interpret its asymptote.
- I can solve work-rate problems.
- I can use inverse variation.

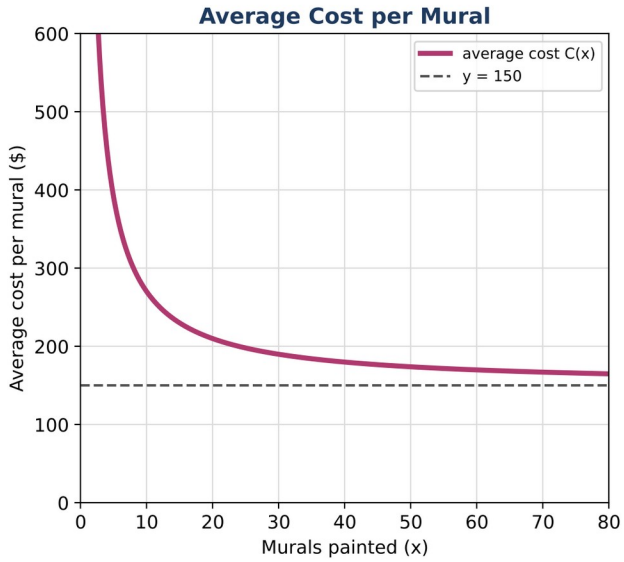
Project Timeline

When	What you will do
Day 1	Average Cost
Day 2	Painters & Rates
Day 3	Business Plan
Final	Business Plan + reflection

Part 1: Average Cost

Day 1 · Average cost per mural: C(x) = (1200 + 150x) ÷ x.

1. Complete the table.



Mural Co.

Equipment $1,200

Materials $150

per mural

Ana: 6 h per mural

Ben: 9 h per mural

Murals x	1	5	10	30	60
C(x) ($)					

2. What value does C(x) get close to as x gets large? Why does that make sense?

3. How many murals must you paint for the average cost to be $190?

4. Why is x = 0 not in the domain?

Part 2: Painters & Rates

Day 2 · Ana paints a mural in 6 hours, so she paints ⅙ of a mural per hour. Ben takes 9 hours.

5. What fraction of a mural does Ben paint per hour?

6. Working together, how long do Ana and Ben take? Solve ⅙ + ⅑ = 1 ÷ t.

7. With Cara, all three finish in 2.4 hours. How long would Cara take alone?

8. A big wall takes 24 painter-days: days = 24 ÷ (painters). How many days with 4 painters? With 6?

Part 3: Business Plan

Day 3 · Plan your first season.

9. Choose a price per mural. How many murals do you need so that the price is at least $50 more than the average cost?

10. Choose a team and find how long one mural takes together.

11. Write a short business plan.

Final Product: Business Plan

Your final product must include:

- Average cost graph with the asymptote labeled
- Team schedule
- Price and goal
- Business plan

Reflection

R1. Why can the average cost never reach $150?

R2. Why do we add rates and not times?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Prices and schedules are based on correctly solved rational equations, with extraneous solutions rejected. Goes beyond the minimum with a creative, realistic choice.	Prices and schedules are based on correctly solved rational equations, with extraneous solutions rejected.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Plan includes the average cost function with asymptotes, work-rate schedules, and a pricing decision supported by equations. Polished, organized, and visually engaging.	Plan includes the average cost function with asymptotes, work-rate schedules, and a pricing decision supported by equations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.